A controlled thermal window for chord generators

Endpoint preparation and the Hamiltonian boost

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Abstract

The zero-chord matter seed fails a direct identification of the Lin–Stanford length-preserving generators with the microscopic boost. We study their exact action on an endpoint Euclidean preparation. A positive-path comparison with the ordinary oscillator gives a sufficient joint window in which the inverse-temperature parameter vanishes, the chord number diverges, and the normalized heat vector approaches a moving coherent vector. In this window the actions of the rescaled chord boost and the negative Hamiltonian boost agree in norm on the prepared vector. Lowest-weight and second-moment limits follow. The preparation becomes orthogonal to its original seed, so the result supplies a controlled moving family, rather than an identification of the original selected state.

1 Conventions and the question being tested

First take the established fixed-λ double-scaled, half-filled limit of the project's complex Gaussian CAR model. The subsequent limit in this note is

$$q = e^{-\epsilon}, \qquad \epsilon = 4\lambda, \qquad K = \frac{H}{2J_s} = \frac{h}{\sqrt{\epsilon}}, \qquad \hbar = 1.$$

In Lin–Stanford (LS) units their deformation parameter is our ϵ, and their Hamiltonian is our K [1, (18)]. We work in the tracial one-insertion chord module of fixed weight $d > 0$, with Hamiltonian–matter crossing $r = q^d$. For the established actual charged curvature channel, $d = 2$; we use that module, without adding an independent random probe. The full selected density is not inserted in this note.

Let $|l, k\rangle = |d; l, k\rangle$, and set $[n] = (1 - q^n)/(1 - q)$. LS creation and annihilation operators give

$$a_L|l, k\rangle = [l]|l - 1, k\rangle + rq^l[k]|l, k - 1\rangle, \tag{1}$$

$$a_R|l, k\rangle = [k]|l, k - 1\rangle + rq^k[l]|l - 1, k\rangle. \tag{2}$$

Creation adds a chord on the indicated side. The total level is $\bar{n} = l+k+d$ and different total levels are orthogonal. We compare $B_{\mathrm{tr}} = (K_L - K_R)/2$ with the LS combinations of $J_{ij} = a_i^\dagger a_j - [\bar{n}]$. The new estimates below use this existing representation and do not constitute a new chord algebra or an independent bulk dual.

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2 A sufficient joint window from positive paths

The endpoint choice of the LS preparation [1, (141)] is

$$\Psi = Z_q^{-1/2} e^{-\beta K_L/2}|0,0\rangle, \qquad t = \frac{\beta}{2\sqrt{\epsilon}}, \quad m = t^2, \quad Z_q = \|e^{-tT_q} e_0\|^2. \tag{3}$$

Here $e_n = |n,0\rangle/\sqrt{[n]!}$ is orthonormal and $T_q = \sqrt{\epsilon} K_L$ has off-diagonal entries $\sqrt{[n]}$. This endpoint subspace is invariant under K_L, but not under the other generators. After a parity transformation the heat vector has positive coefficients $x = e^{tT_q} e_0$. The comparison vector $y = e^{tT_1} e_0$ satisfies

$$0 \le x_n \le y_n, \qquad \|y\|^2 = e^{2m}, \qquad \chi_n = (-1)^n e^{-m/2} \frac{m^{n/2}}{\sqrt{n!}}.$$

Thus $|\chi_n|^2$ is a Poisson law of mean m.

Proposition 1. *For $\delta = 2\epsilon m^2 < 1$,*

$$1 - \delta \le Z_q / e^{2m} \le 1, \tag{4}$$

$$\|\Psi - \chi\|^2 \le 2(1 - \sqrt{1-\delta}) \le 2\delta, \tag{5}$$

$$|\Psi_n|^2 \le \frac{e^{-m} m^n}{(1-\delta) n!}, \qquad |\Psi_0| \le \frac{e^{-m/2}}{\sqrt{1-\delta}}. \tag{6}$$

The identification in (5) uses the orthonormal endpoint basis for each q, not the raw chord basis.

Proof. The $2j$-th vacuum moment of T_q sums q^{cr} over pairings. Each has at most $j(j-1)/2$ crossings, so its deficit from the $q = 1$ weight is at most $\epsilon j(j-1)/2$. In the normalized series for $\langle e_0, e^{2tT_1} e_0\rangle$, j has law Pois$(2m)$. Its factorial moment gives total deficit at most $2\epsilon m^2$. Positivity gives $\langle x, y\rangle \ge \|x\|^2$, proving (5). The componentwise bound and (4) give (6); for the last bound use $y_0 = e^{m/2}$. □

The sufficient, nonempty window is

$$m \longrightarrow \infty, \quad \epsilon m^2 \longrightarrow 0, \qquad \text{equivalently} \qquad \sqrt{\epsilon} \ll \beta \ll \epsilon^{1/4}. \tag{7}$$

In particular, $\mathbb{E}_\Psi (N/m - 1)^2 \le [(1-\delta)m]^{-1}$. Thus $N/m \to 1$, $\epsilon N \to 0$, and the seed overlap tends to zero. This is a sufficient window, not an optimal one. The fixed-temperature LS saddle motivates it but does not by itself prove this joint limiting statement.

3 Exact generator action at the endpoint

In the sector $l+k=n$, put $c_n = q^{(n+d)/2}$ and use the exact LS definitions [1, (48), (147)]:

$$E = -\frac{J_{LL}+J_{RR}}{2c_n}, \qquad B = \frac{J_{LL}-J_{RR}}{2c_n}, \qquad P = \frac{i}{2}(J_{LR}-J_{RL}), \tag{8}$$

$$\widehat{B} = \frac{B}{\sqrt{1-c_n^2}}, \qquad \widehat{P} = \frac{P}{\sqrt{1-c_n^2}}. \tag{9}$$

The rescaling is sectorwise. All these operators preserve n. For $n \geq 1$, define

$$u_n = \frac{|n-1,1\rangle}{\sqrt{[n]!}}, \qquad \langle e_n, u_n\rangle = r, \qquad \|u_n\|^2 = r^2 + \frac{1-r^2}{[n]}, \qquad f_n = \frac{u_n - r e_n}{\sqrt{(1-r^2)/[n]}}.$$

The Gram identities follow by moving a creation operator to the bra and applying its annihilation adjoint. They make e_n, f_n orthonormal, avoiding subtraction of nearly parallel raw vectors in the numerical implementation. Write $g_n = \sqrt{[n](1-r^2)}$. Direct application of the annihilators gives

$$\widehat{B}e_n = \frac{[n](1-r^2)e_n - r g_n f_n}{2c_n\sqrt{1-c_n^2}}, \tag{10}$$

$$\widehat{P}e_n = -\frac{i g_n f_n}{2\sqrt{1-c_n^2}}, \tag{11}$$

$$Ee_n = \frac{[d]-(1-r)^2[n]/2}{c_n} e_n - \frac{r g_n}{2c_n} f_n. \tag{12}$$

At $n=0$ the f_n terms are absent, $B=P=0$ and $E=[d]/q^{d/2}$, recovering the previously recorded seed obstruction.

For an arbitrary endpoint vector $\psi = \sum_n \psi_n e_n$, the level-n components of the physical boost are instead

$$(B_{\mathrm{tr}}\psi)_{e,n} = \frac{1-r}{2\sqrt{\epsilon}}\left(\sqrt{[n]}\,\psi_{n-1} + \sqrt{[n+1]}\,\psi_{n+1}\right), \tag{13}$$

$$(B_{\mathrm{tr}}\psi)_{f,n} = -\frac{\sqrt{1-r^2}}{2\sqrt{\epsilon}}\,\psi_{n-1}. \tag{14}$$

Missing coefficients are zero. Equations (13)–(14) retain both lowering and raising, including the first output level beyond a numerical cutoff. The comparison is between operators in the same chord Hilbert space, rather than between two independently fitted correlation functions.

4 Matching on the moving preparation

Proposition 2. *For each fixed $d > 0$, in (7),*

$$\|(\widehat{B} + B_{\mathrm{tr}})\Psi\| \longrightarrow 0, \qquad \|(E - d)\Psi\| \longrightarrow 0, \qquad \|(\widehat{B} + i\widehat{P})\Psi\| \longrightarrow 0, \tag{15}$$

$$\langle \widehat{B} \rangle_\Psi,\ \langle B_{\mathrm{tr}} \rangle_\Psi \longrightarrow 0, \qquad \langle \widehat{B}^2 \rangle_\Psi,\ \langle \widehat{P}^2 \rangle_\Psi, \langle B_{\mathrm{tr}}^2 \rangle_\Psi \longrightarrow d/2. \tag{16}$$

Also $\langle [\widehat{B}, \widehat{P}] \rangle_\Psi \to id$. These are one-vector and weak-matrix-element limits; no convergence of commutators in norm or of the full operators is asserted.

Proof. First work on the comparison vector χ. On levels with $n \to \infty$ and $\epsilon n \to 0$,

$$[n]/n \to 1, \quad (1 - r^2)/\epsilon \to 2d, \quad (1 - c_n^2)/(\epsilon(n + d)) \to 1.$$

Equations (10)–(12) therefore give, uniformly in the Poisson bulk as $n/m \to 1$,

$$\widehat{B} e_n = -\sqrt{d/2}\, f_n + o(1), \qquad \widehat{P} e_n = -i\sqrt{d/2}\, f_n + o(1), \qquad E e_n = d e_n + o(1).$$

Here vector remainders are in the level's orthonormal coordinates. The coherent coefficients satisfy exactly $\chi_{n-1} = -\sqrt{n/m}\,\chi_n$ and $\chi_{n+1} = -\sqrt{m/(n+1)}\,\chi_n$. Consequently (14) approaches $+\sqrt{d/2}\,\chi_n f_n$, while (13) vanishes: its bulk size is $O_d(\sqrt{\epsilon m})$. This proves the plus sign in (15).

To justify the bulk restriction and transfer from χ to Ψ, one needs more than norm closeness alone. The squared endpoint coefficients of $\widehat{B}, \widehat{P}, E$, and the two shifted coefficients of B_{tr}, obey a common envelope

$$C_d(1 + \epsilon(n + 1))^2 e^{2\epsilon(n+d+1)}, \qquad 0 < \epsilon \le 1. \tag{17}$$

For example, use $[n] \le n$, $1 - r^2 \le 2d\epsilon$ and $1 - e^{-z} \ge z e^{-z}$; the diagonal part of $\widehat{B}$ squared is at most $d^2 \epsilon n e^{2\epsilon(n+d)}$, and its f_n part squared at most $(d/2)e^{2\epsilon(n+d)}$. The other coefficients follow from the same inequalities. On $m/2 \le n \le 2m$ the envelope is bounded uniformly. On its complement, the Poisson domination (6) and exponential tilting of $\mathrm{Pois}(m)$ make the weighted tail tend to zero. Indeed, tilting by $e^{2\epsilon n}$ gives mean $m e^{2\epsilon}$ and prefactor $e^{m(e^{2\epsilon}-1)} \to 1$; the fixed polynomial weights do not overcome the exponential concentration. Shifts by one level have the same property. Equation (5) now transfers each graph norm. The moments follow by inner products; specifically $\langle [\widehat{B}, \widehat{P}] \rangle = 2i\,\mathrm{Im}\langle \widehat{B}\Psi, \widehat{P}\Psi \rangle \to id$. □

5 Finite checks, truncations, and numerical corrections

The exact rational checks use $q = 1/4, 4/9$, $d = 1, 2, 3$ and levels $0 \le n \le 7$. They test the full chord Gram symmetry and creation/annihilation adjoint relation, endpoint Gram and generator formulas, and both boost degree shifts. Independent pairing enumeration through order twelve checks the positive vacuum-moment input. There are 4,152 exact equalities in total.

For the thermal vectors we use $m = 8, 32, 128, 512$, $\epsilon = m^{-3}$, $\beta = 2/m$, so $\delta = 2/m$. The coarse Jacobi cutoffs are $122, 226, 530, 1522$; each is increased by 64. The scaled positive Taylor vector is $e^{-m}\sum_{j=0}^{L}(tT_{q,M})^j e_0/j!$. The actual Taylor orders are $274, 570, 1451, 3628$.

Here are bounds that avoid numerical underflow. Let $\pi_\mu(k) = \Pr\{\mathrm{Pois}(\mu) \ge k\}$. Positive paths and their maximum level give, in exact arithmetic,

$$\|\Psi - \Psi_M\|^2 \le \frac{2\pi_{2m}(M+1)}{1-\delta}, \tag{18}$$

$$\left\| e^{-m} e^{tT_{q,M}} e_0 - e^{-m} \sum_{j=0}^{L} \frac{(tT_{q,M})^j}{j!} e_0 \right\|^2 \le \pi_{2m}(\lceil (L+1)/2 \rceil). \tag{19}$$

For (18), a vacuum path visiting level $M+1$ has at least $M+1$ up steps. For (19), positivity bounds the squared remainder by the difference of squared norms. Missing pairs of Taylor powers have even total degree $2j \ge L+1$. In both cases compare to the $q = 1$ moments. We evaluate the tails using $\log \pi_\mu(k) \le -\mu + k(1 + \log(\mu/k))$, $k > \mu$. The largest vector bounds are 1.68×10^{-23} for the Jacobi cutoff and 8.41×10^{-39} for the scaled Taylor tail.

m	$\langle N \rangle$	$\|(\widehat{B} + B_{\mathrm{tr}})\Psi\|$	$\|(E-2)\Psi\|$	$\langle \widehat{B}^2 \rangle$
8	7.897924	0.171789	0.125648	0.837580
32	31.973991	0.087559	0.031260	0.943929
128	127.993490	0.044095	0.007813	0.984796
512	511.998372	0.022085	0.001953	0.996120

These entries use the actual channel $d = 2$; the same driver also checks $d = 1, 3$. At $m = 512$, $\langle B_{\mathrm{tr}}^2 \rangle = 1.000015$, whereas $\|(\widehat{B} - B_{\mathrm{tr}})\Psi\| = 1.997945$, confirming the sign. The cancellation ratio $(\|\widehat{B}\Psi\| + \|B_{\mathrm{tr}}\Psi\|)/\|(\widehat{B} + B_{\mathrm{tr}})\Psi\|$ is at most 90.48 for this channel. It is recorded rather than suppressed.

6 What the comparison establishes and what comes next

The single-step matching and the limiting lowest-weight relations provide a concrete route beyond the old zero-chord test. They do not leave the preparation fixed. In fact, (6) proves $\langle e_0, \Psi\rangle \to 0$; at $m = 512$ the overlap is about 6.62×10^{-112}. The same bulk/tail argument, applied to $T_q/\sqrt{m}$, gives

$$\|(T_q/\sqrt{m} + 2)\Psi\| \longrightarrow 0.$$

Thus the thermal vector moves far in rms Hamiltonian units even though $\beta \to 0$. A claim that this regulator simply restores the original trace preparation would be incorrect.

The original full selected preparation has

$$\rho = 1 - \sqrt{q}(1 + \lambda) + \sqrt{q}(h - \sqrt{\lambda}U)^2.$$

Its U dependence is not represented by the endpoint heat vector above. A useful next step is an explicit moving-space identification, followed by mixed generator matrix elements with this density retained. A displacement on the endpoint oscillator alone has not yet been extended to the entire matter module. Products of generators, commutator norms on descendants, and the selected mixed Ward identities remain to be established. Neither $d = 2$ nor $E\Psi \simeq 2\Psi$ fixes a Poincare mass; positive null translations and their multiparticle composition still require a separate construction. The selected four-bilocal and finite-G dynamics are also outside this result.

Numerical record. Long calls to `expm_multiply` failed the initial precision check; splitting and normalizing reduced, but did not eliminate, the discrepancy. The largest split-method vector difference is 3.45×10^{-9}, and this method is retained only as a diagnostic. The positive Taylor calculation was checked at 50 decimal digits: the largest relative discrepancy among the eight recorded $d = 2$ observables is 3.16×10^{-11}, in the small coherent-distance quantity. The initial `longdouble` comparison did not increase precision on this Windows host; the supplemental decimal check explicitly corrects that description. An underflowed zero in the original Taylor-bound field is likewise superseded by (19) and the separate logarithmic tail audit. The bounds are for exact arithmetic, not interval certificates for all floating-point operations. Failed diagnostics and both corrections are preserved alongside the sealed scientific outputs.

Predecessors and attribution. The q-oscillator, one-matter representation, Euclidean preparation and rescaled generators are LS inputs. Central-spectrum coherent behavior and thermal chord wavefunctions are already studied by Xu [2, Secs. 2.3, 5]; q-coherent preparations are also discussed in [3, Sec. 3.2.1]. Our ordinary exponential heat vector is not defined to be a q-annihilation eigenstate. This note supplies a sufficient norm-controlled joint window and an endpoint application of the existing generators. We do not claim priority for coherent-state methods or for emergent $\mathfrak{sl}_2$ symmetry.

References

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